PulseIQ Analytics - Phase 1 Report

Data Analysts Track | Days 1-5 | Prepared 23 October 2026

Name	Poonam Satish Patole	**Phase**	Phase 1
GitHub repository	PoonamPatole/pulseiq	**Git tag**	phase-1

Five-task review

Task	Status	Evidence / outcome
1. ETL and data contracts	Complete locally	Four supplied feeds use detailed contracts; dictionary type/nullability checks cover remaining facts. Row-count, ETL, quarantine and rerun evidence are documented.
2. Star-schema warehouse	Complete locally	11 dimensions and 12 facts; row-count reconciliation and 261 explained relationship exceptions are documented.
3. Data-quality gatekeeper	Complete locally	14 checks run; scratch self-tests detect injected defects; 11 planted issue categories were detected.
4. Governed metric layer	Complete locally	28 supplied metrics plus M29, catalog, privacy thresholds and five hand checks.
5. Workforce dashboard	Complete locally	Streamlit dashboard with filters, four KPI cards, four charts, CEO Questions page and screenshots.

Data-quality note. The supplied data currently produces five Critical Task 3 findings. The gate records these after loading; affected metrics remain provisional until findings are resolved or formally accepted.

Proof it works

? All 29 CSV files match DuckDB raw-table row counts; rerun verification found materialized raw, clean, summary and quarantine tables stable.

? All 12 warehouse fact tables reconcile by row count; Task 1 quarantines 20 invalid fact_task rows with reasons.

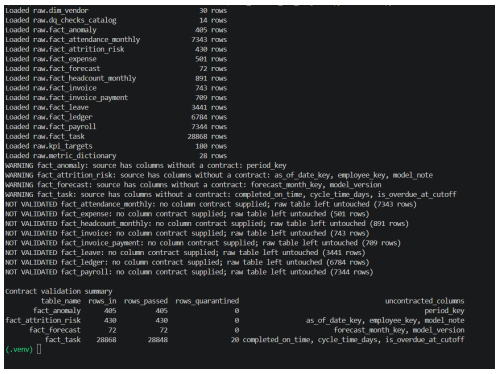
? Dashboard AppTest reports zero exceptions, four KPI cards, four charts, department and date filters. September 2026 headcount is 441; M29 growth is +3.28%.

? M28 forecast error is NULL with NO_ACTUALS_YET because actual headcount does not cover forecast months.

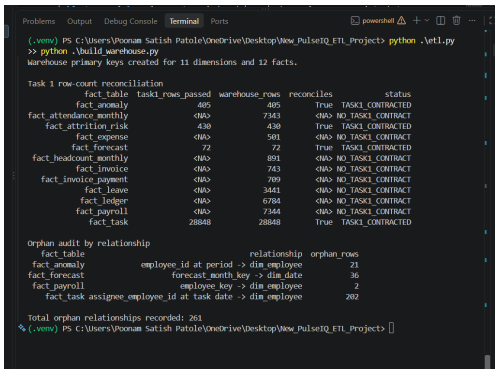
Evidence and checks

Selected screenshots from Tasks 1, 2 and 5

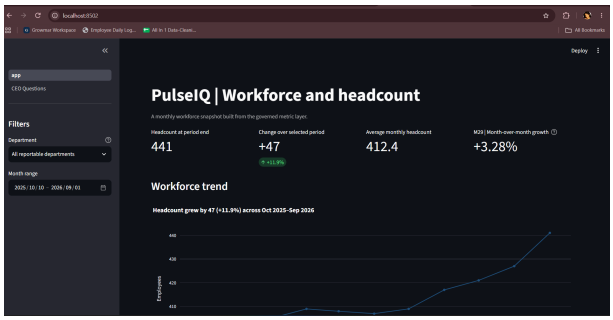
Task 1: CSV and raw-table count evidence



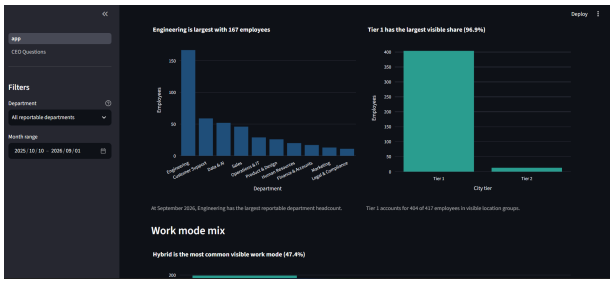
Task 2: warehouse build and audit evidence



Task 5: dashboard KPIs, filters and trend



Task 5: department and workforce composition



Mini-challenge: M29 headcount growth month over month

M29 = (current period-end headcount ? previous period-end headcount) / previous period-end headcount ? 100. Grain: company-month or department-month. Groups below five are suppressed.

Month	Current	Previous	Hand calculation	View
Sep 2026	441	427	(441 ? 427) / 427 ? 100 = 3.278689%	+3.28%

Task 4 calculator checks

Metric	Hand calculation	Result
M01 Headcount	D01 Sep 2026: period-end headcount	167
M04 Attrition	2 exits / 160 starting headcount ? 100	1.25%
M07 Gross pay	D01 Sep 2026 sum across 169 employees	INR 20,720,901.38
M16 Revenue	Account 4001, Apr 2025: negative signed ledger sum	INR 41,824,400
M22 Task completion	T01 / SP-01: 50 done / 55 non-cancelled	90.91%

Reflection and next steps

PulseIQ Analytics | Phase 1

What I learned

DuckDB stores an analytics database in one local file and can load CSV data directly. A star schema keeps measurable events in fact tables and descriptive context in dimensions. Employee surrogate keys preserve which department was valid when a historical event happened. Data contracts and quality checks make failures visible. Metric views give dashboards reusable definitions for business numbers.

Problems and solutions

Detailed business rules were supplied for four feeds; dictionary type and nullability rules cover remaining fact columns without inventing unsupported rules. Numeric-looking text identifiers are converted only in clean tables. Quality findings and relationship exceptions remain documented for investigation instead of being silently removed.

Plan for Phase 2

Resolve or formally accept Critical data-quality findings with documented owners. Then extend the dashboard to attendance, attrition, teams and payroll cost, using governed metric views and preserving small-group privacy thresholds.

Remaining submission steps

1. Add the programme GitHub account **fobesitech** as a collaborator.

2. Review the report, export the final PDF and upload it to the programme site.

Project and version

Repository: https://github.com/PoonamPatole/pulseiq
Git tag: **phase-1**
Prepared: **23 October 2026**